

K-Subgraph Isomorphism Program

Compilation Guide (Windows)

Mateusz Osik, Mateusz Nędzi, Mateusz Polis

December 10, 2025

1 Project Structure

```
k-subgraph_isomorphism/  
  Makefile.windows  
  include/  
    approx.h  
    exact.h  
    io.h  
  src/  
    main.cpp  
    approx.cpp  
    exact.cpp  
    io.cpp  
  examples/  
    *.txt
```

2 Compilation

Open a terminal and navigate to the project root directory. Execute:

```
make -f Makefile.windows
```

This produces the standalone executable `aac.exe`.

2.1 Build Output

Artifact	Location
Object files	<code>build_win/</code>
Executable	<code>aac.exe</code> (project root)

2.2 Clean Build

To remove all compiled files and rebuild from scratch:

```
make -f Makefile.windows clean  
make -f Makefile.windows
```

3 Compiler Configuration

The `Makefile.windows` uses the following settings:

Setting	Value / Purpose
Compiler	g++ (MinGW-w64)
-std=c++20	C++20 standard
-Wall	Enable all common warnings
-Iinclude	Header search path
-static-libgcc	Statically link GCC runtime
-static-libstdc++	Statically link C++ standard library

The static linking flags ensure `aac.exe` runs independently without requiring external runtime DLLs.

4 Verification

After successful compilation:

```
aac.exe exact 3 examples/example1.txt
```

The program should execute and produce output without errors.